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SAMI RAUTANEN

AI ENGINEER & AGENTIC SYSTEMS
ARCHITECT



PROFILE

I am an **AI Engineer** specializing in autonomous multi-agent systems and AWS serverless cloud infrastructure. I design, build, and **ship production-grade AI systems** — from architecture through CI/CD pipelines to live deployments. My work spans multi-agent orchestration (CrewAI, LangGraph, OpenAI Agents SDK), AWS infrastructure managed with Terraform, and full-stack development with React and FastAPI. I don't just prototype — I deploy.



KEY PROJECTS

AgentSquad Platform

React · FastAPI · OpenAI SDK · Resend

Full-stack multi-agent platform — [Live on Vercel](#)

- Built a **6-agent Sales Intelligence pipeline** that drafts, evaluates, and sends real emails via Resend with domain authentication (SPF/DKIM).
- Developed a **parallel Deep Research team** executing 3 simultaneous searches and synthesizing cited executive reports.
- Deployed React frontend on Vercel, FastAPI backend on Hugging Face Spaces with rate limiting and PIN authentication.

Digital Twin — AWS Production

AWS Lambda · Terraform · Bedrock
· FastAPI

Serverless AI deployment on AWS — [Live on CloudFront](#)

- Deployed FastAPI backend as AWS Lambda function via Mangum adapter with API Gateway and CloudFront CDN.
- Managed full AWS infrastructure with **Terraform IaC** (multi-environment: dev/test/prod) and GitHub Actions CI/CD using OIDC keyless authentication.
- Integrated Amazon Bedrock (Nova) models replacing OpenAI, with S3-based conversation memory.

EngineeringTeam Crew

CrewAI · Claude 3.7 · GPT-4o · Gradio

Autonomous 5-agent software team — [Live on Hugging Face](#)

📍 Finland

✉ samrude1@outlook.com

🔗 LinkedIn Profile

📁 GitHub (27 Repos)

🌐 samirautanen.fi

EXPERTISE

AI Agents (CrewAI / LangGraph)



AWS Serverless (Lambda / Bedrock)



Terraform / IaC / CI/CD



Python / FastAPI



React / Next.js / TypeScript



LLM Integration (GPT / Claude / Gemini)



EDUCATION

C# Systems Development

Taitotalo Helsinki 2023 - Software
Training

Technical Design

Hyria Ammattiopisto Vocational
Qualification

LANGUAGES

Finnish (Native)



English (Pro Read/Write)



Working Spoken Proficiency

- Orchestrated a sequential crew: Lead Architect → Backend Engineer → Frontend Engineer → QA Engineer → Technical Writer.
- Delivers complete Python applications (design doc, backend code, Gradio UI, Pytest suite, README) in under 10 minutes.
- Implemented rate limiting (15 runs/IP), input validation, and concurrency queuing for production safety.

Sidekick AI Agent

LangGraph · Gemini 2.5 Flash · Playwright

Autonomous tool-using agent — [Live on Hugging Face](#)

- Implemented **Worker-Evaluator pattern** in LangGraph: agent executes tasks, evaluator assesses output and triggers self-correction loops.
- Integrated real browser control via Playwright and dynamic code execution via Python REPL as agent tools.
- Anti-abuse measures: Gradio concurrency limits, LangGraph recursion cap (8), protecting API budget in public deployment.



ADDITIONAL EXPERIENCE

- **Game Development & Algorithms:** Released 8 titles on itch.io.
 - *Tech:* Javascript/HTML5 (Canvas API, Custom Physics), Unity/C#, WebAssembly (WASM).
 - *Key Projects:* **Chess Terminal** (Stockfish integration), **CellWars** (Simulation logic), and custom physics engines.
- **3D & Technical Art:** Background in 3D visualization and CAD design, bringing strong spatial reasoning to system architecture.
- **Full-Stack Web Apps:** *TreeniTrack* (AI Coach integration) and *Activity Finder* (Geospatial React App).



PROFESSIONAL HISTORY

Murata Electronics Oy (formerly VTI Technologies)

1999 -
2024

Senior Production Operator

- 25-year career in high-precision manufacturing (MEMS technology) for automotive and medical sectors.
- Disciplined operation in **ISO 4/5 Cleanroom** environments, ensuring strict quality control and adherence to technical protocols.